

Analytical Geometry

1. Coordinate geometry

- Cartesian coordinates (x, y) to polar coordinates (r, θ) :

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \text{in quadrant I}$$

$$= \pi - \tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in quadrant II}$$

$$= \pi + \tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in quadrant III}$$

$$= -\tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in quadrant IV}$$

- Polar coordinates (r, θ) to Cartesian coordinates (x, y) :

$$x = r \cos \theta$$

$$y = r \sin \theta$$

- Distance between two points in Cartesian coordinates $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Distance between two points in polar coordinates $P(r_1, \theta_1)$ and $Q(r_2, \theta_2)$:

$$d = \sqrt{r_1^2 + r_2^2 - 2r_1r_2 \cos(\theta_1 - \theta_2)}$$

- Section formula (internal division):

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

- Section formula (external division):

$$x = \frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \quad y = \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2}$$

- Midpoint formula:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

- Area of triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$\Delta ABC = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

- Centroid G of triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

- Incenter I of triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and side lengths a, b, c :

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right)$$

- Excenter E_A opposite to vertex A :

$$E_A = \left(\frac{-ax_1 + bx_2 + cx_3}{-a + b + c}, \frac{-ay_1 + by_2 + cy_3}{-a + b + c} \right)$$

- Circumcenter O of triangle ABC :

$$O = \left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

- Orthocenter H of triangle ABC :

$$H = \left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C} \right)$$

2. Straight lines

- General form of a straight line:

$$ax + by + c = 0$$

- Slope (m) of a line passing through $(x_1, y_1), (x_2, y_2)$, making an angle θ with the positive x-axis:

$$m = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$$

- Slope-intercept form (slope m , y-intercept c):

$$y = mx + c$$

- Point-slope form (passes through (x_1, y_1) with slope m):

$$y - y_1 = m(x - x_1)$$

- Two-point form (passes through (x_1, y_1) and (x_2, y_2)):

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

- Intercept form (x-intercept a , y-intercept b):

$$\frac{x}{a} + \frac{y}{b} = 1$$

- Normal (perpendicular) form (perpendicular distance from origin p , angle α with positive x-axis):

$$x \cos \alpha + y \sin \alpha = p$$

- Parametric / symmetric form (distance r from (x_1, y_1) along angle θ):

$$\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be coincident:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be parallel to each other:

$$m_1 = m_2 \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

- Equation of a line parallel to $ax + by + c = 0$:

$$ax + by + k = 0$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be perpendicular to each other:

$$m_1m_2 = -1 \Rightarrow a_1a_2 + b_1b_2 = 0$$

- Equation of a line orthogonal to $ax + by + c = 0$:

$$bx - ay + k = 0$$

- Angle between two lines with slopes m_1 and m_2 :

$$\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1m_2}$$

- Perpendicular distance from (x_1, y_1) to $ax + by + c = 0$:

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

- Perpendicular distance between two parallel lines $ax + by + c_1 = 0$, $ax + by + c_2 = 0$:

$$d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

- Foot of perpendicular (h, k) from (x_1, y_1) to $ax + by + c = 0$:

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$$

- Image / reflection (h, k) of (x_1, y_1) in line $ax + by + c = 0$:

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -2\frac{ax_1 + by_1 + c}{a^2 + b^2}$$

- Equation of a line that bisects the angle between two intersecting lines $(a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0)$:

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

1. If $a_1a_2 + b_1b_2 > 0$:

- The + sign gives the obtuse angle bisector.

- The - sign gives the acute angle bisector.
- 2. If $a_1a_2 + b_1b_2 < 0$:
 - The + sign gives the acute angle bisector.
 - The - sign gives the obtuse angle bisector.
- Family of lines passing through the intersection of $L_1 = 0$ and $L_2 = 0$:

$$L_1 + \lambda L_2 = 0$$

- Conditions for concurrency of three lines ($a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0, a_3x + b_3y + c_3 = 0$):

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

- Angle between a pair of straight lines passing through the origin ($ax^2 + 2hxy + by^2 = 0$):

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}$$

3. Conics

- General equation of a conic:

$$ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$$

If $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$, then it represents a pair of straight lines

If $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \neq 0$,

$$\Rightarrow a = b, h = 0 \rightarrow \text{Circle}$$

$$\Rightarrow ab - h^2 = 0 \rightarrow \text{Parabola}$$

$$\Rightarrow ab - h^2 > 0 \rightarrow \text{Ellipse}$$

$$\Rightarrow ab - h^2 < 0 \rightarrow \text{Hyperbola}$$

3.1. Circle

- General equation of a circle:

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

- Center: $(-g, -f)$
 - Radius: $\sqrt{g^2 + f^2 - c}$
 - Length of x-axis intercept: $2\sqrt{g^2 - c}$
 - Length of y-axis intercept: $2\sqrt{f^2 - c}$
 - Touches x-axis: $g^2 = c$
 - Touches y-axis: $f^2 = c$
 - Touches both axes: $g^2 = f^2 = c$
 - Center on x-axis: $f = 0$
 - Center on y-axis: $g = 0$
 - Center at origin: $g = 0$ and $f = 0$
 - Passes through origin: $c = 0$
 - Touches x-axis at origin: $g = 0$ and $c = 0$
 - Touches y-axis at origin: $f = 0$ and $c = 0$
- Central form:

$$(x - h)^2 + (y - k)^2 = r^2 \quad \text{where center} = (h, k) \text{ and radius} = r$$

- Parametric forms:

- For $x^2 + y^2 = r^2$:

$$x = r \cos(\theta), \quad y = r \sin(\theta) \quad (0 \leq \theta < 2\pi)$$

- For $(x - h)^2 + (y - k)^2 = r^2$:

$$x = h + r \cos(\theta), \quad y = k + r \sin(\theta)$$

- Diameter form (endpoints of diameter are (x_1, y_1) and (x_2, y_2)):

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

- Position of a point (x_1, y_1) relative to $S = x^2 + y^2 + 2gx + 2fy + c = 0$:

$$\text{Let } S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$$

- Outside: $S_1 > 0$
- On circumference: $S_1 = 0$
- Inside: $S_1 < 0$

- Condition for tangency (line $y = mx + c$ to circle $x^2 + y^2 = r^2$):

$$c = \pm r\sqrt{1 + m^2}$$

- Tangent in slope form:

$$y = mx \pm r\sqrt{1 + m^2}$$

- Point of contact:

$$\left(\mp \frac{rm}{\sqrt{1 + m^2}}, \pm \frac{r}{\sqrt{1 + m^2}} \right)$$

- Equation of tangent at point (x_1, y_1) on circle $S = 0$:

$$T = 0 \quad \text{where } T = xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$$

- Equation of normal at point (x_1, y_1) :

- For $x^2 + y^2 = r^2$:

$$yx_1 - xy_1 = 0$$

- For $x^2 + y^2 + 2gx + 2fy + c = 0$:

$$\frac{x - x_1}{x_1 + g} = \frac{y - y_1}{y_1 + f}$$

- Length of tangent from external point (x_1, y_1) :

$$L = \sqrt{S_1}$$

- Angle θ between tangents drawn from external point (x_1, y_1) :

$$\tan\left(\frac{\theta}{2}\right) = \frac{r}{\sqrt{S_1}}$$

- Pair of tangents from external point (x_1, y_1) :

$$S \cdot S_1 = T^2$$

- Chord of contact from external point (x_1, y_1) :

$$T = 0$$

- Chord bisected at given point (x_1, y_1) :

$$T = S_1$$

- Area of the triangle formed by tangents and chord of contact given external point (x_1, y_1) and radius r :

$$\Delta = \frac{r \cdot S_1^{\frac{3}{2}}}{S_1 + r^2}$$

- Director circle (locus of perpendicular tangents):
 - For circle $x^2 + y^2 = r^2$:

$$x^2 + y^2 = 2r^2$$

- Family of circles through intersections of circles $C_1 = 0$ and $C_2 = 0$:

$$C_1 + kC_2 = 0 \quad (k \neq -1)$$

- Family of circles through intersections of circle $C = 0$ and line $L = 0$:

$$C + kL = 0$$

- Equation of co-axial family of circles:

$$S + \lambda L = 0 \text{ or } S_1 + \lambda(S_1 - S_2) = 0$$

- Common chord of two circles $S = 0$ and $S' = 0$:

$$S - S' = 0$$

- Radical axis of two circles $S = 0$ and $S' = 0$:

$$S - S' = 0$$

- Radical center of three circles $S_1 = 0$, $S_2 = 0$, and $S_3 = 0$ taken two at a time:

$$S_1 - S_2 = 0 \text{ and } S_2 - S_3 = 0 \text{ and } S_3 - S_1 = 0$$

- Angle of intersection between circles C_1 and C_2 :

$$\cos(\theta) = \frac{r_1^2 + r_2^2 - d^2}{2r_1r_2} = \frac{2(g_1g_2 + f_1f_2) - (c_1 + c_2)}{2\sqrt{g_1^2 + f_1^2 - c_1}\sqrt{g_2^2 + f_2^2 - c_2}}$$

- Orthogonal intersection ($\theta = 90^\circ$):

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

- Relative position (centers C_1, C_2 , radii r_1, r_2 , center distance $d = C_1C_2$):

- One inside another (no intersection): $d < |r_1 - r_2|$
- Touch internally (1 common tangent): $d = |r_1 - r_2|$
- Intersect at two points (2 common tangents): $|r_1 - r_2| < d < r_1 + r_2$

- Touch externally (3 common tangents): $d = r_1 + r_2$
- Completely separate (4 common tangents): $d > r_1 + r_2$
- Length of common tangents:
 - Direct common tangent: $L_d = \sqrt{d^2 - (r_1 - r_2)^2}$
 - Transverse common tangent: $L_t = \sqrt{d^2 - (r_1 + r_2)^2}$
- Equation of a circle through 3 non-collinear points:

$$\begin{vmatrix} x^2 + y^2 & x & y & 1 \\ x_1^2 + y_1^2 & x_1 & y_1 & 1 \\ x_2^2 + y_2^2 & x_2 & y_2 & 1 \\ x_3^2 + y_3^2 & x_3 & y_3 & 1 \end{vmatrix} = 0$$

- Polar of a point (x_1, y_1) with respect to $S = 0$:

$$T = 0$$

- Pole of a line $lx + my + n = 0$ with respect to $x^2 + y^2 = r^2$:

$$\left(-\frac{r^2 l}{n}, -\frac{r^2 m}{n} \right)$$

- Points dividing the line joining centers internally and externally in the ratio of radii $(r_1 : r_2)$:
 - Internal center of similitude:

$$\left(\frac{r_1 x_2 + r_2 x_1}{r_1 + r_2}, \frac{r_1 y_2 + r_2 y_1}{r_1 + r_2} \right)$$

- External center of similitude:

$$\left(\frac{r_1 x_2 - r_2 x_1}{r_1 - r_2}, \frac{r_1 y_2 - r_2 y_1}{r_1 - r_2} \right)$$